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## ICE 2.3



Aman Gupta • Feb 5, 2025 (Edited Feb 15, 2025)

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1. (2 marks) What is the time complexity of launching a brute-force attack on polyalphabetic cipher where the key is known to be a string of 5 English lower-case letters. Assume that the plaintext is itself a string of lower-case English letter.

- $O(5!)$
- $O(26!)$
- **$O(26^5)$  (correct)**
- $O(5^{26})$
- None of the above

2. (3 marks) What is the time complexity of launching a brute-force attack on transposition cipher (discussed in class) where the key length is known to be a string of 5 digits. Assume that the plaintext is itself a string of lower case English letter.

- **$O(5!)$  (correct)**
- $O(26!)$
- $O(26^5)$
- $O(5^{26})$
- None of the above

3. (3 marks) Continuing with question 2 above, in what way does the time complexity of a brute-force attack change if there was to be TWO rounds of transposition encipherment. Assume that key length, and not the key itself, is the same for round 1 and for round 2, viz 5. And the time complexity of either round is  $O(f(5, 26))$

- $O(f(5, 26)) + O(f(5, 26))$
- **$O(f(5, 26)) * O(f(5, 26))$  (correct)**
- $O(f(10, 26))$
- None of the above

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## Your work

Graded

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